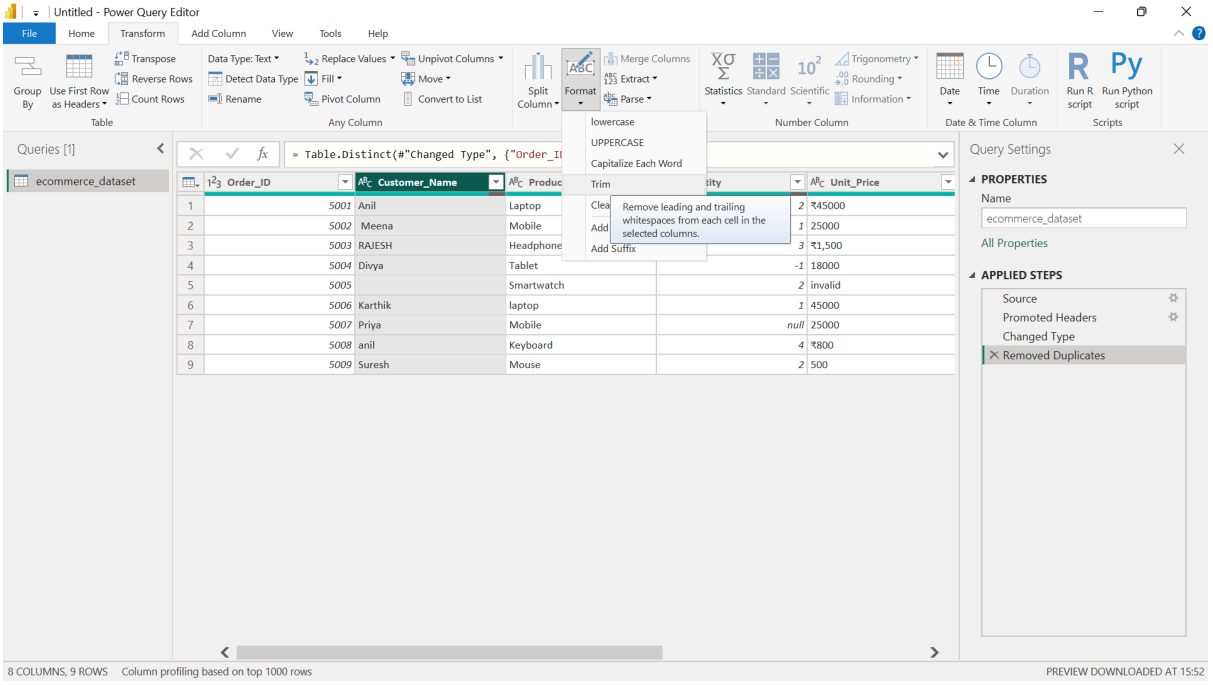
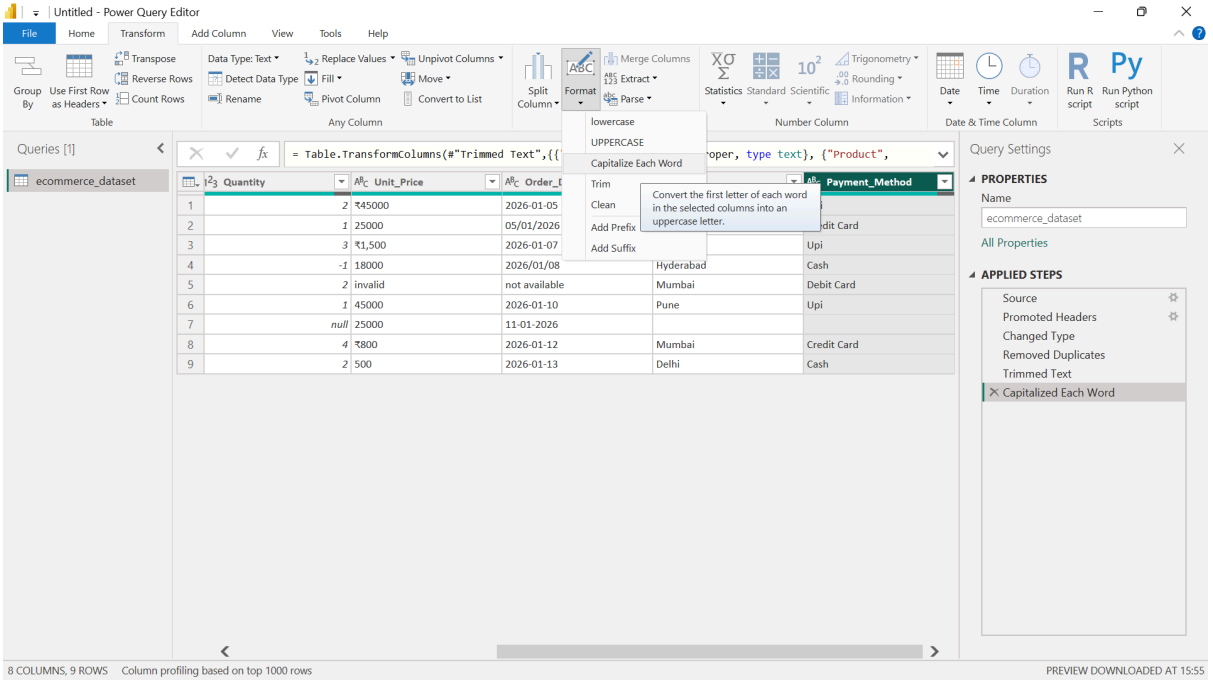
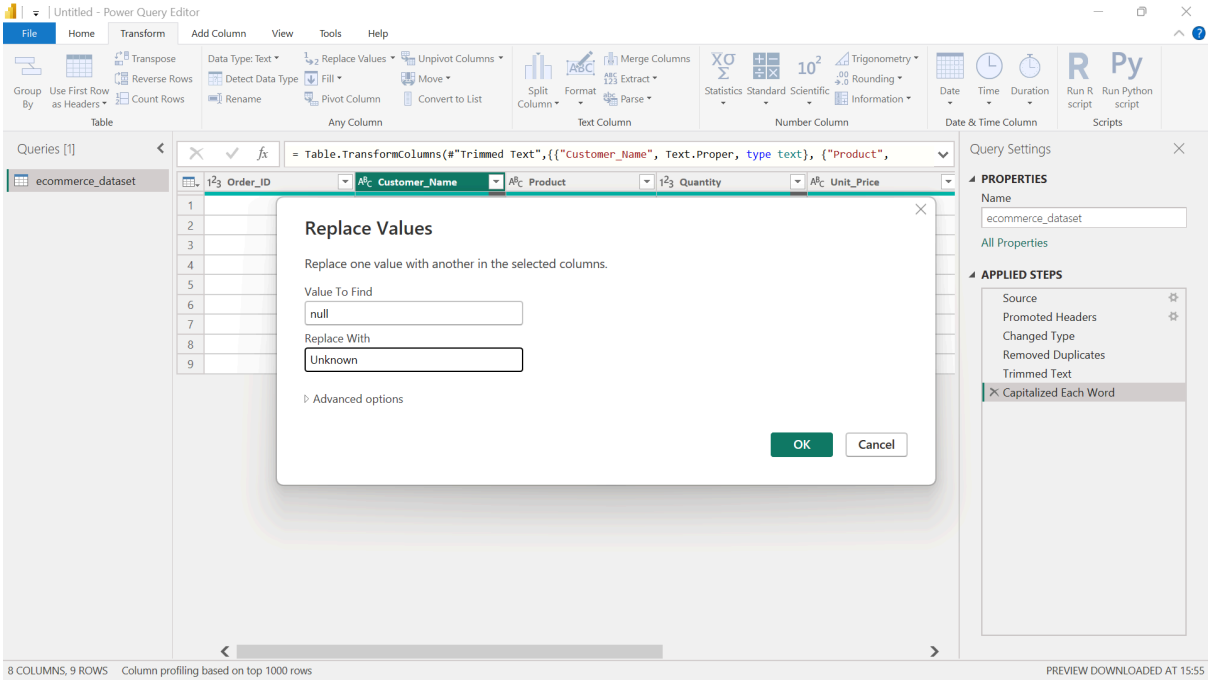
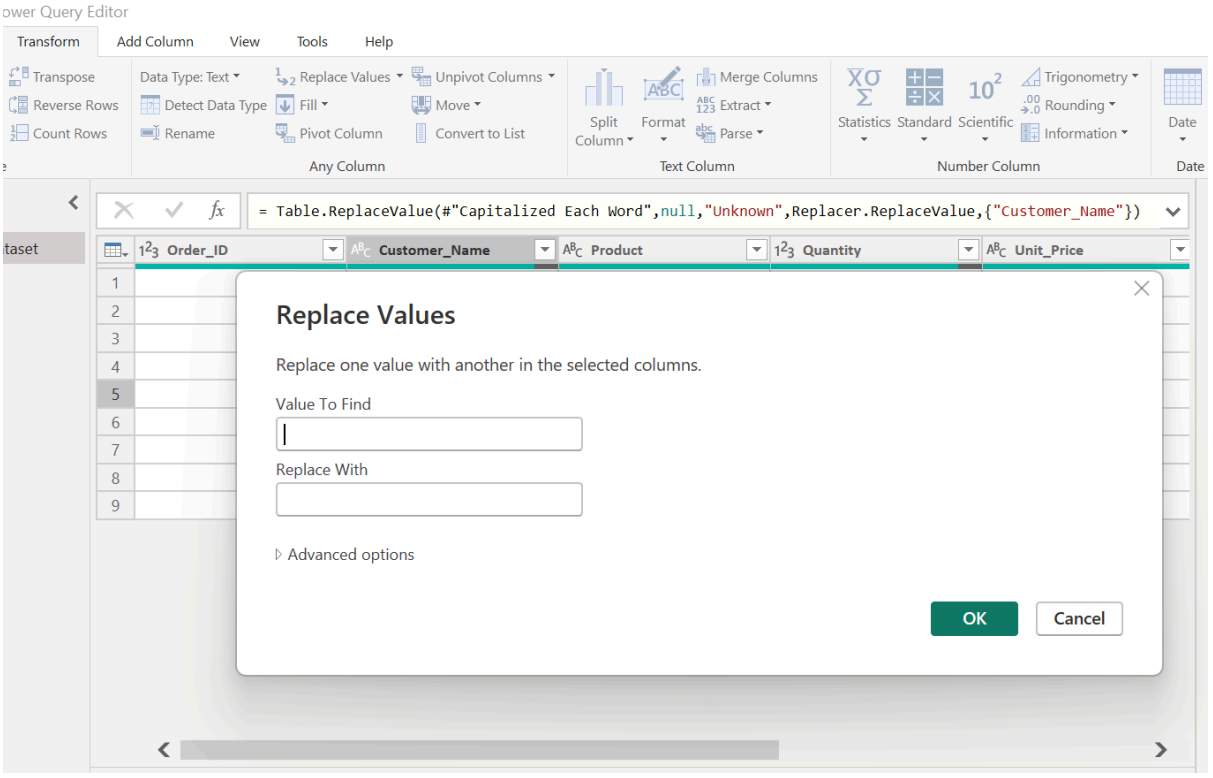
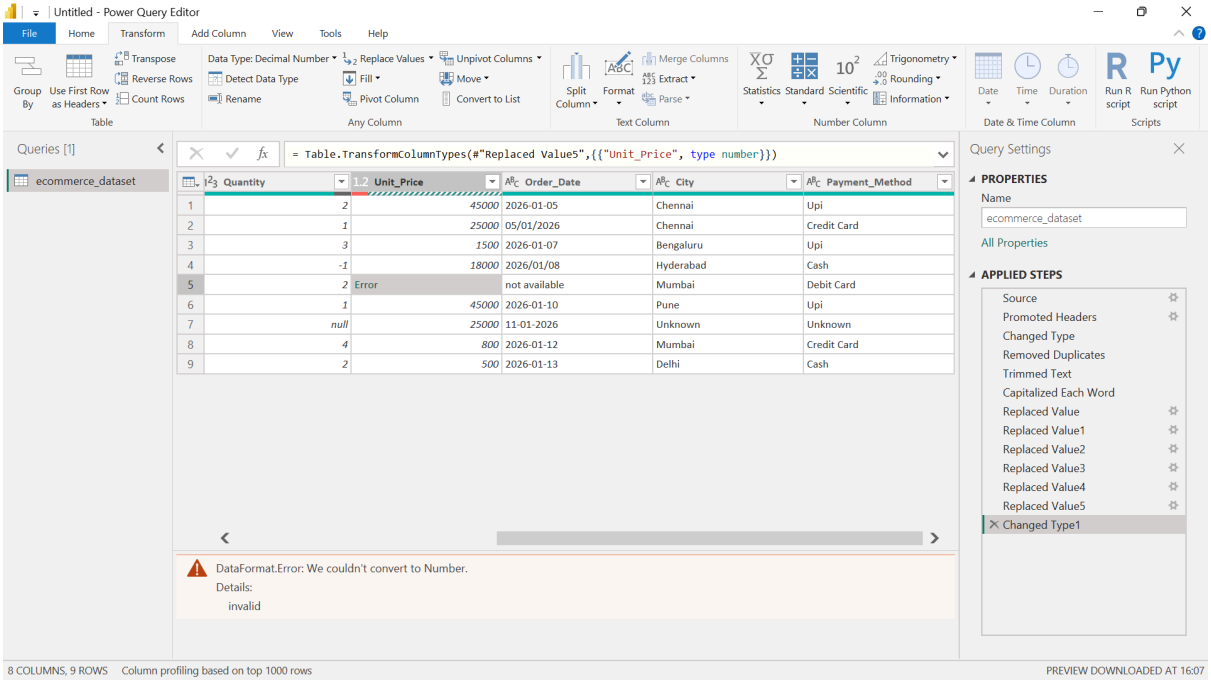
Data Loading, Transformation, Relationships, and Visualization Using Power BI

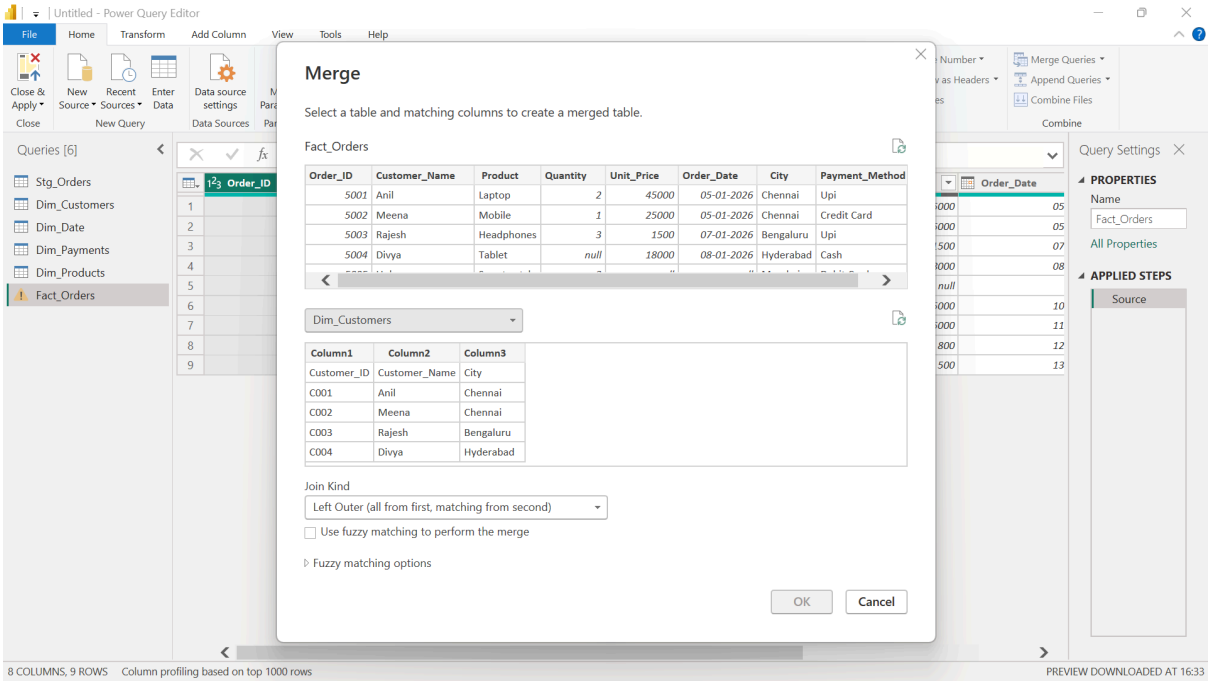


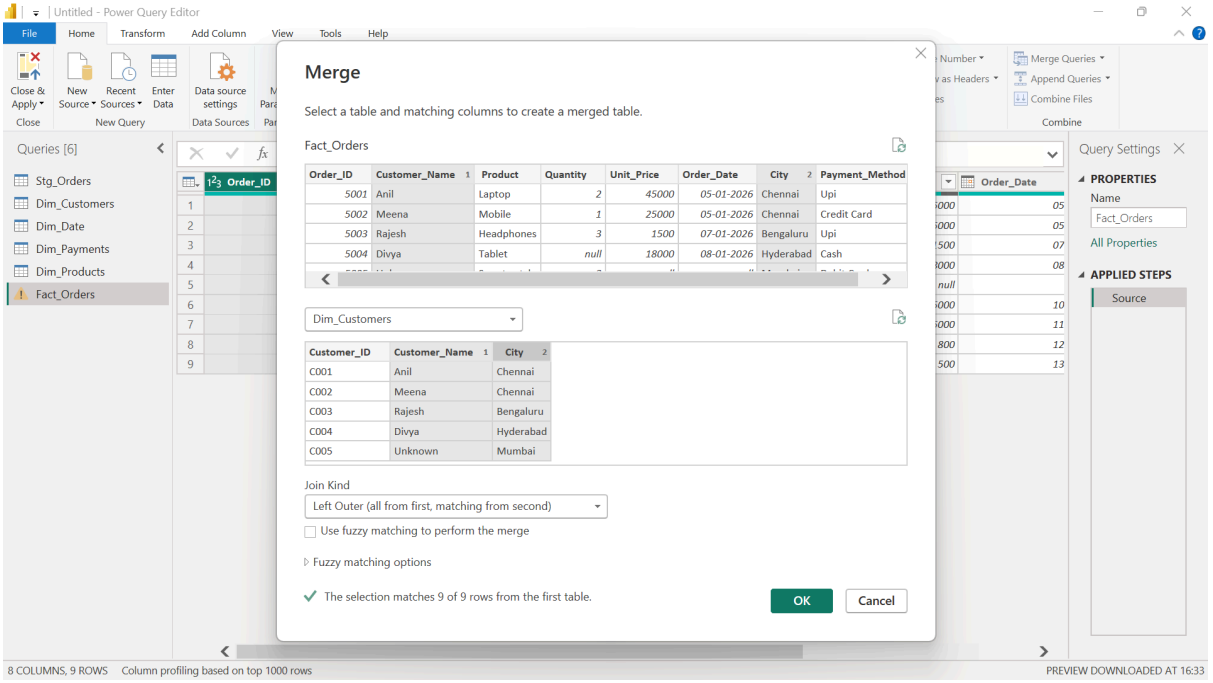


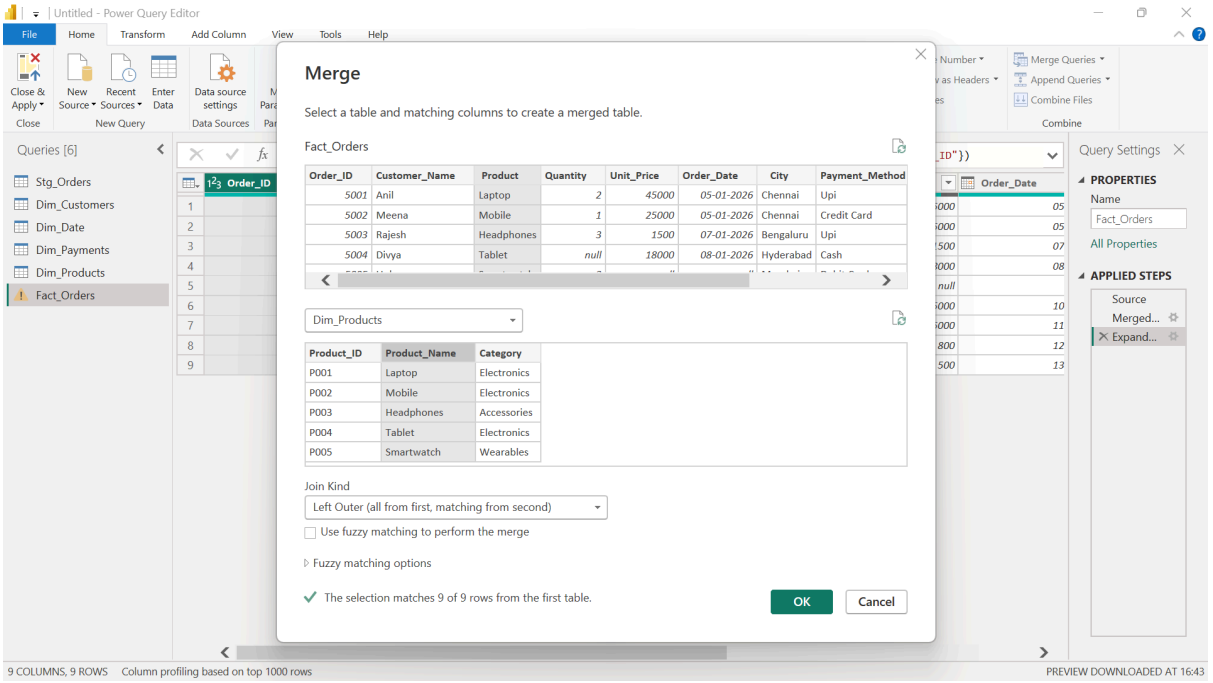

Untitled - Power Query Editor
File
Home
Transform
Add Column
View
Tools
Help
Queries [1]
ecommerce_dataset
= Table.TransformColumns(#"Trimmed Text",{{"Customer_Name", Text.Proper, type text}, {"Product",
Order_ID
Customer_Name
Product
Quantity
Unit_Price
Replace Values
Replace one value with another in the selected columns.
Value To Find
null
Replace With
Unknown
Advanced options
OK
Cancel
Query Settings
PROPERTIES
Name
ecommerce_dataset
All Properties
APPLIED STEPS
Source
Promoted Headers
Changed Type
Removed Duplicates
Trimmed Text
Capitalized Each Word
8 COLUMNS, 9 ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 15:55

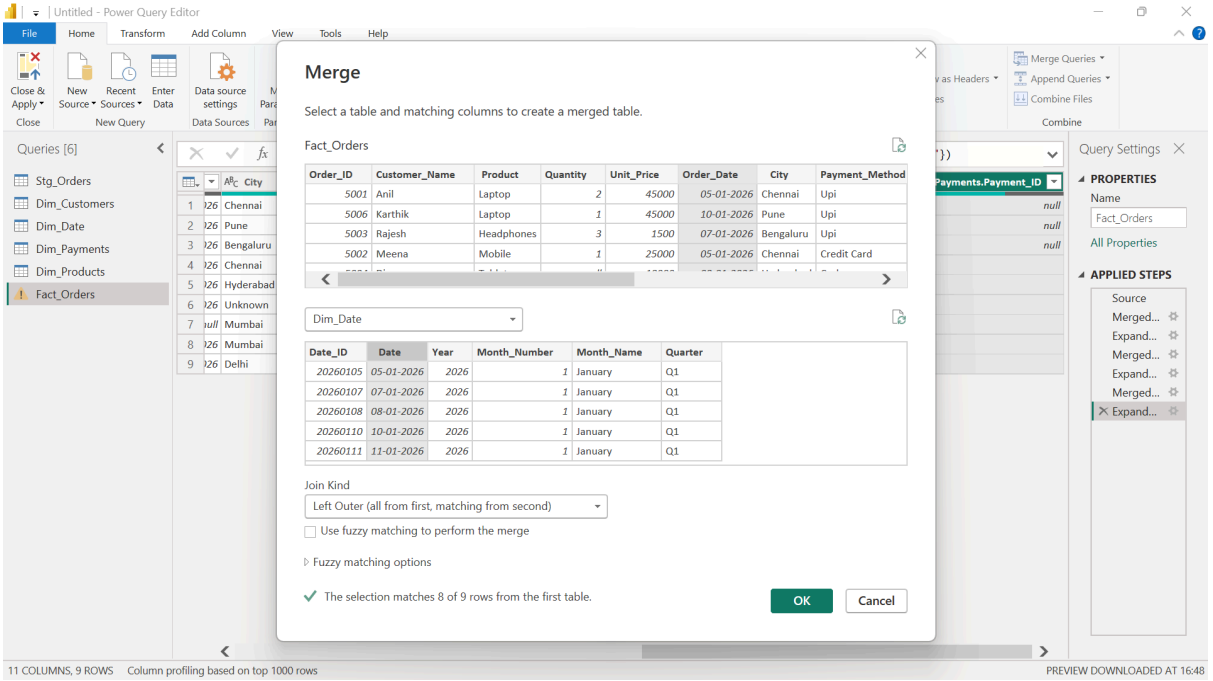

ower Query Editor
Transform
Add Column
View
Tools
Help
Transpose
Reverse Rows
Count Rows
Data Type: Text
Detect Data Type
Rename
Replace Values
Fill
Pivot Column
Unpivot Columns
Move
Convert to List
Any Column
Split Column
Format
Merge Columns
Extract
Parse
Text Column
Statistics
Standard
Scientific
Trigonometry
Rounding
Information
Number Column
Date
= Table.ReplaceValue(#"Capitalized Each Word",null,"Unknown",Replacer.ReplaceValue,{"Customer_Name"})
Order_ID
Customer_Name
Product
Quantity
Unit_Price
Replace Values
Replace one value with another in the selected columns.
Value To Find
Replace With
Advanced options
OK
Cancel

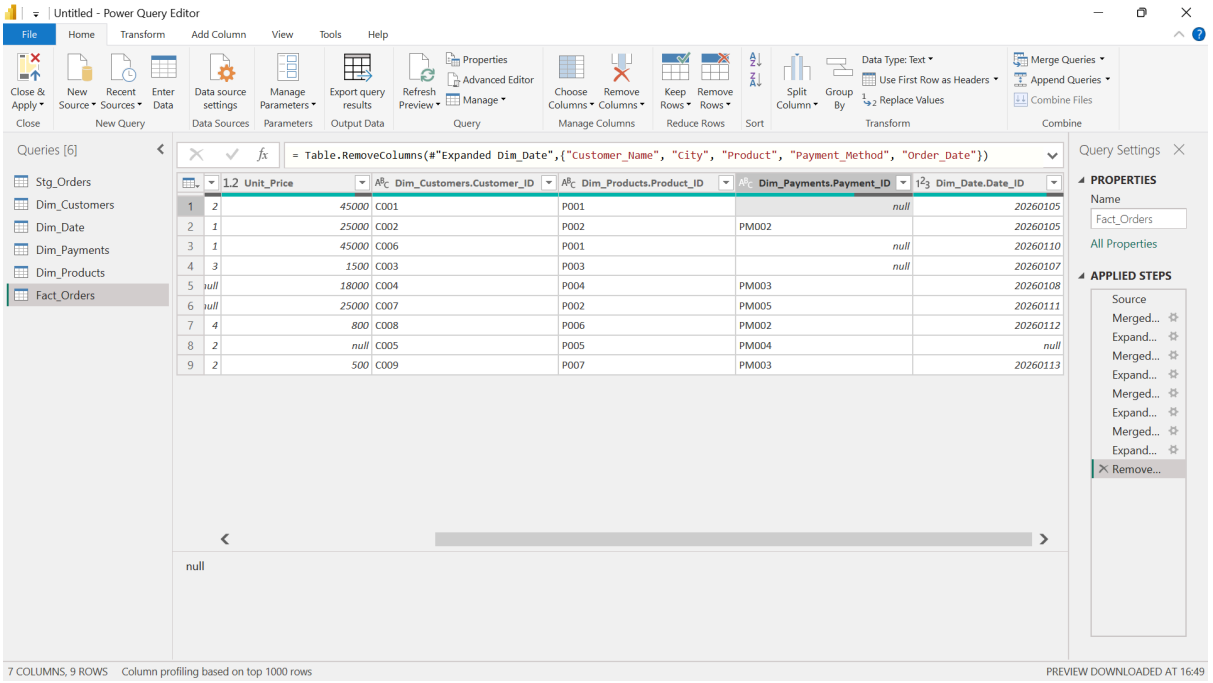

Untitled - Power Query Editor
File
Home
Transform
Add Column
View
Tools
Help
Queries [1]
ecommerce_dataset
= Table.TransformColumnTypes(#"Replaced Value5",{{"Unit_Price", type number}})
Quantity
Unit_Price
Order_Date
City
Payment_Method
1 2 45000 2026-01-05 Chennai Upi
2 1 25000 05/01/2026 Chennai Credit Card
3 3 1500 2026-01-07 Bengaluru Upi
4 -1 18000 2026/01/08 Hyderabad Cash
5 2 Error not available Mumbai Debit Card
6 1 45000 2026-01-10 Pune Upi
7 null 25000 11-01-2026 Unknown Unknown
8 4 800 2026-01-12 Mumbai Credit Card
9 2 500 2026-01-13 Delhi Cash
DataFormat.Error: We couldn't convert to Number.
Details:
invalid
Query Settings
PROPERTIES
Name
ecommerce_dataset
All Properties
APPLIED STEPS
Source
Promoted Headers
Changed Type
Removed Duplicates
Trimmed Text
Capitalized Each Word
Replaced Value
Replaced Value1
Replaced Value2
Replaced Value3
Replaced Value4
Replaced Value5
Changed Type1
8 COLUMNS, 9 ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 16:07

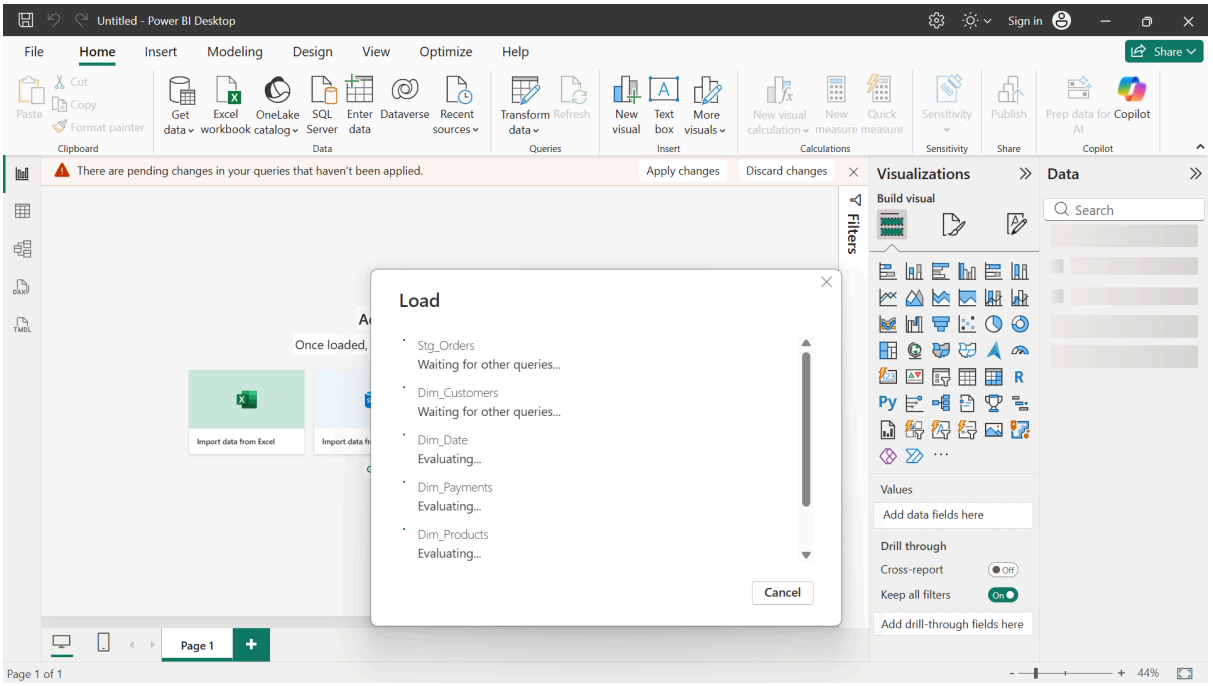

Untitled - Power Query Editor
Merge
Select a table and matching columns to create a merged table.
Fact_Orders
Order_ID Customer_Name Product Quantity Unit_Price Order_Date City Payment_Method
5001 Anil Laptop 2 45000 05-01-2026 Chennai Upi
5002 Meena Mobile 1 25000 05-01-2026 Chennai Credit Card
5003 Rajesh Headphones 3 1500 07-01-2026 Bengaluru Upi
5004 Divya Tablet null 18000 08-01-2026 Hyderabad Cash
Dim_Customers
Column1 Column2 Column3
Customer_ID Customer_Name City
C001 Anil Chennai
C002 Meena Chennai
C003 Rajesh Bengaluru
C004 Divya Hyderabad
Join Kind
Left Outer (all from first, matching from second)
Use fuzzy matching to perform the merge
Fuzzy matching options
OK
Cancel
Queries [6]
Stg_Orders
Dim_Customers
Dim_Date
Dim_Payments
Dim_Products
Fact_Orders
Query Settings
PROPERTIES
Name
Fact_Orders
All Properties
APPLIED STEPS
Source
8 COLUMNS, 9 ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 16:33

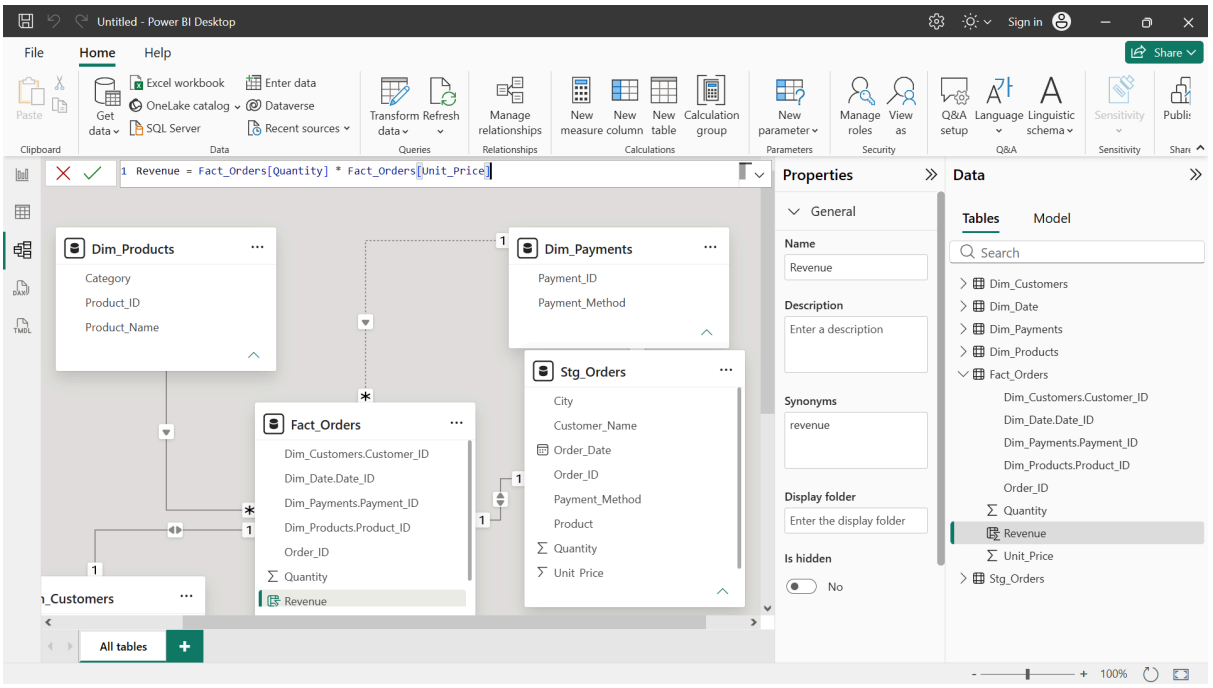

Untitled - Power Query Editor
Merge
Select a table and matching columns to create a merged table.
Fact_Orders
Order_ID | Customer_Name | Product | Quantity | Unit_Price | Order_Date | City | Payment_Method
5001 | Anil | Laptop | 2 | 45000 | 05-01-2026 | Chennai | Upi
5002 | Meena | Mobile | 1 | 25000 | 05-01-2026 | Chennai | Credit Card
5003 | Rajesh | Headphones | 3 | 1500 | 07-01-2026 | Bengaluru | Upi
5004 | Divya | Tablet | null | 18000 | 08-01-2026 | Hyderabad | Cash
Dim_Customers
Customer_ID | Customer_Name | City
C001 | Anil | Chennai
C002 | Meena | Chennai
C003 | Rajesh | Bengaluru
C004 | Divya | Hyderabad
C005 | Unknown | Mumbai
Join Kind
Left Outer (all from first, matching from second)
Use fuzzy matching to perform the merge
Fuzzy matching options
The selection matches 9 of 9 rows from the first table.
OK
Cancel
Queries [6]
Stg_Orders
Dim_Customers
Dim_Date
Dim_Payments
Dim_Products
Fact_Orders
Query Settings
PROPERTIES
Name
Fact_Orders
All Properties
APPLIED STEPS
Source
8 COLUMNS, 9 ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 16:33

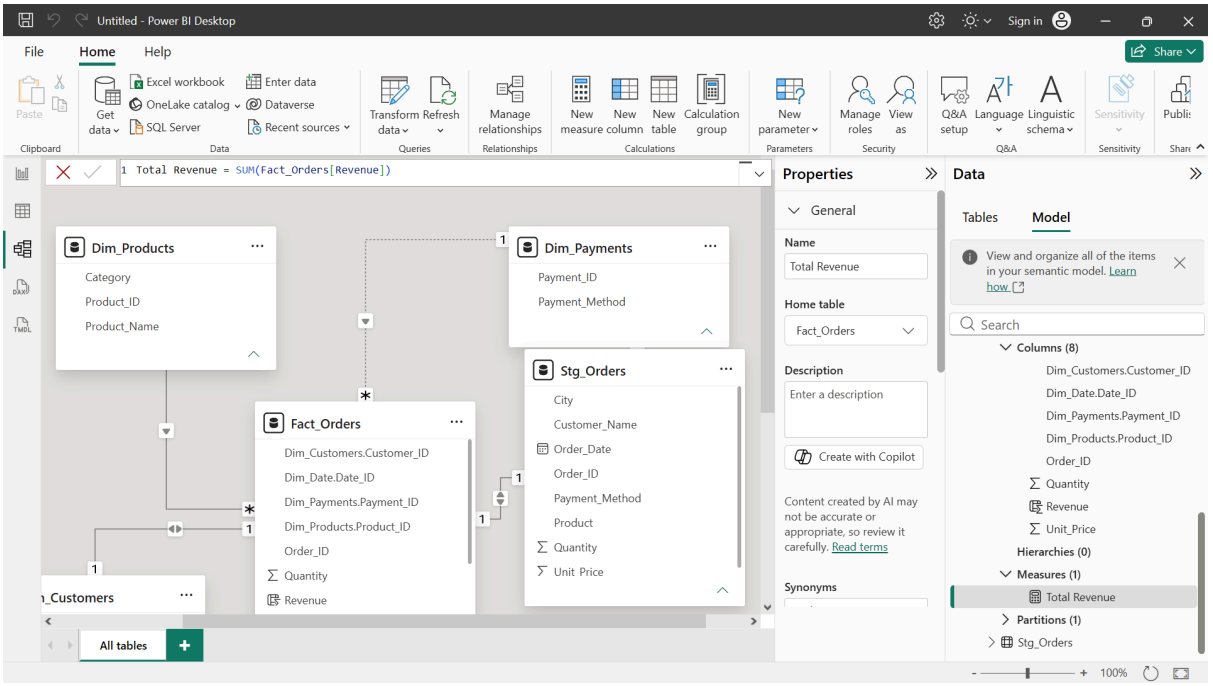

Untitled - Power Query Editor
Merge
Select a table and matching columns to create a merged table.
Fact_Orders
Order_ID | Customer_Name | Product | Quantity | Unit_Price | Order_Date | City | Payment_Method
5001 | Anil | Laptop | 2 | 45000 | 05-01-2026 | Chennai | Upi
5002 | Meena | Mobile | 1 | 25000 | 05-01-2026 | Chennai | Credit Card
5003 | Rajesh | Headphones | 3 | 1500 | 07-01-2026 | Bengaluru | Upi
5004 | Divya | Tablet | null | 18000 | 08-01-2026 | Hyderabad | Cash
Dim_Products
Product_ID | Product_Name | Category
P001 | Laptop | Electronics
P002 | Mobile | Electronics
P003 | Headphones | Accessories
P004 | Tablet | Electronics
P005 | Smartwatch | Wearables
Join Kind
Left Outer (all from first, matching from second)
Use fuzzy matching to perform the merge
Fuzzy matching options
The selection matches 9 of 9 rows from the first table.
OK
Cancel
Queries [6]
Stg_Orders
Dim_Customers
Dim_Date
Dim_Payments
Dim_Products
Fact_Orders
Query Settings
PROPERTIES
Name
Fact_Orders
All Properties
APPLIED STEPS
Source
Merged...
Expand...
9 COLUMNS, 9 ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 16:43

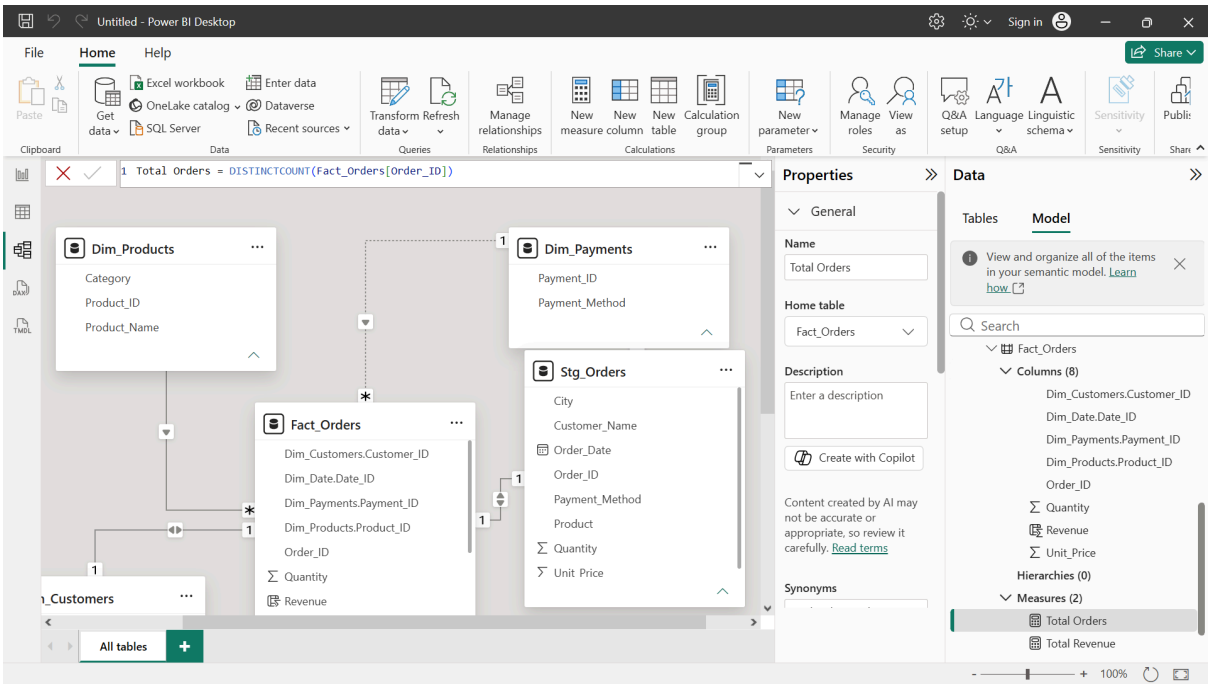

Untitled - Power Query Editor
Merge
Select a table and matching columns to create a merged table.
Fact_Orders
Order_ID | Customer_Name | Product | Quantity | Unit_Price | Order_Date | City | Payment_Method
5001 | Anil | Laptop | 2 | 45000 | 05-01-2026 | Chennai | Upi
5006 | Karthik | Laptop | 1 | 45000 | 10-01-2026 | Pune | Upi
5003 | Rajesh | Headphones | 3 | 1500 | 07-01-2026 | Bengaluru | Upi
5002 | Meena | Mobile | 1 | 25000 | 05-01-2026 | Chennai | Credit Card
Dim_Date
Date_ID | Date | Year | Month_Number | Month_Name | Quarter
20260105 | 05-01-2026 | 2026 | 1 | January | Q1
20260107 | 07-01-2026 | 2026 | 1 | January | Q1
20260108 | 08-01-2026 | 2026 | 1 | January | Q1
20260110 | 10-01-2026 | 2026 | 1 | January | Q1
20260111 | 11-01-2026 | 2026 | 1 | January | Q1
Join Kind
Left Outer (all from first, matching from second)
Use fuzzy matching to perform the merge
Fuzzy matching options
The selection matches 8 of 9 rows from the first table.
OK
Cancel
Queries [6]
Stg_Orders
Dim_Customers
Dim_Date
Dim_Payments
Dim_Products
Fact_Orders
Query Settings
PROPERTIES
Name
Fact_Orders
All Properties
APPLIED STEPS
Source
Merged...
Expand...
Merged...
Expand...
Merged...
Expand...
11 COLUMNS, 9 ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 16:48

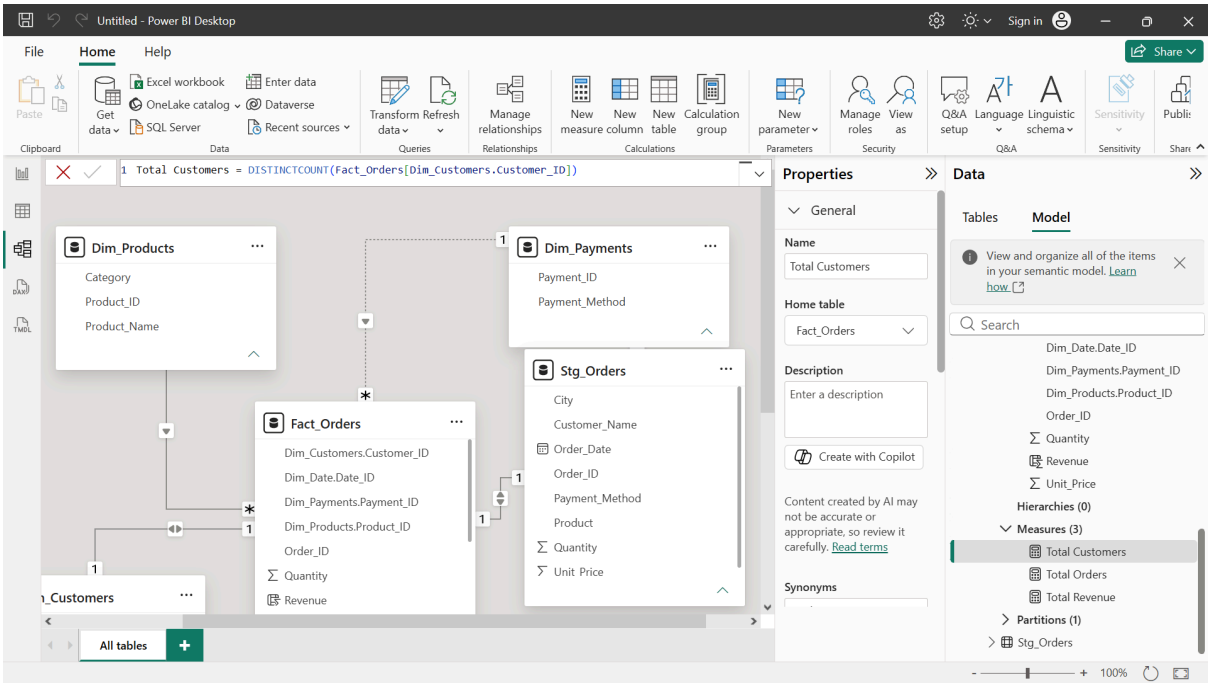

Untitled - Power Query Editor
= Table.RemoveColumns(#"Expanded Dim_Date",{"customer_Name", "City", "Product", "Payment_Method", "Order_Date"})
Unit_Price | Dim_Customers.Customer_ID | Dim_Products.Product_ID | Dim_Payments.Payment_ID | Dim_Date.Date_ID
2 | 45000 | C001 | P001 | null | 20260105
1 | 25000 | C002 | P002 | PM002 | 20260105
1 | 45000 | C006 | P001 | null | 20260110
3 | 1500 | C003 | P003 | null | 20260107
null | 18000 | C004 | P004 | PM003 | 20260108
null | 25000 | C007 | P002 | PM005 | 20260111
4 | 800 | C008 | P006 | PM002 | 20260112
2 | null | C005 | P005 | PM004 | null
2 | 500 | C009 | P007 | PM003 | 20260113
null
Queries [6]
Stg_Orders
Dim_Customers
Dim_Date
Dim_Payments
Dim_Products
Fact_Orders
Query Settings
PROPERTIES
Name
Fact_Orders
All Properties
APPLIED STEPS
Source
Merged...
Expand...
Merged...
Expand...
Merged...
Expand...
Merged...
Expand...
Remove...
7 COLUMNS, 9 ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 16:49

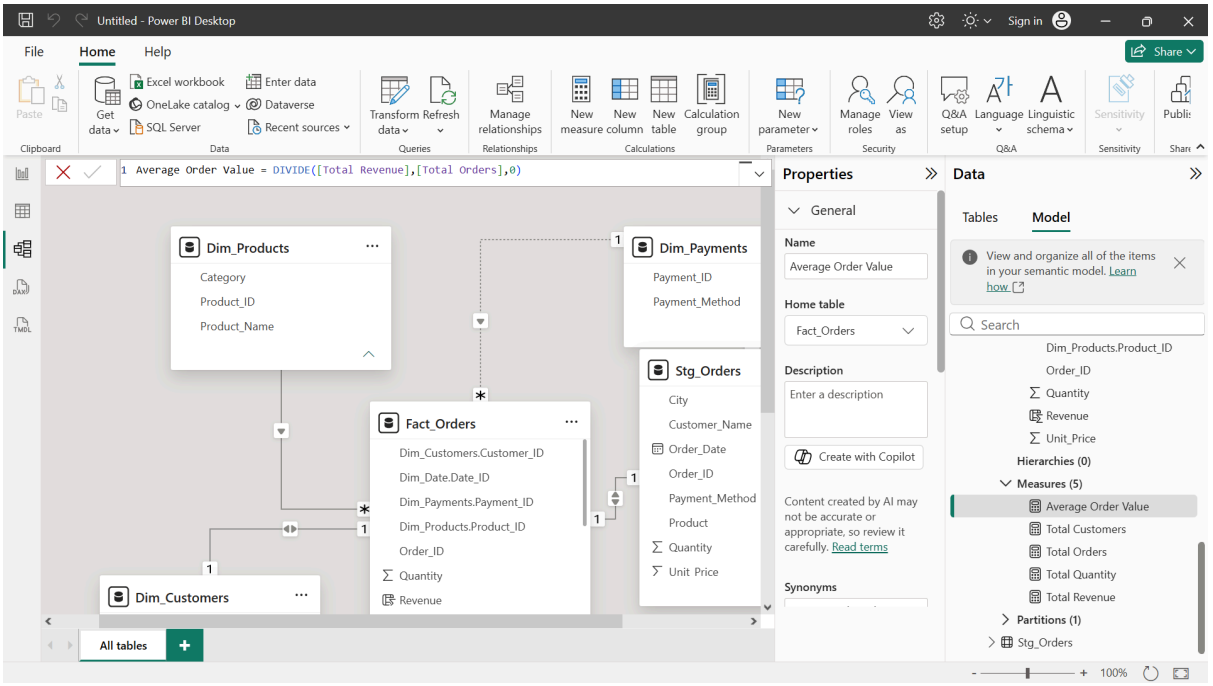

Untitled - Power BI Desktop
Sign in
File
Home
Insert
Modeling
Design
View
Optimize
Help
Share
There are pending changes in your queries that haven't been applied.
Apply changes
Discard changes
Load
Stg_Orders
Waiting for other queries...
Dim_Customers
Waiting for other queries...
Dim_Date
Evaluating...
Dim_Payments
Evaluating...
Dim_Products
Evaluating...
Cancel
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Data
Search
Page 1
Page 1 of 1
44%

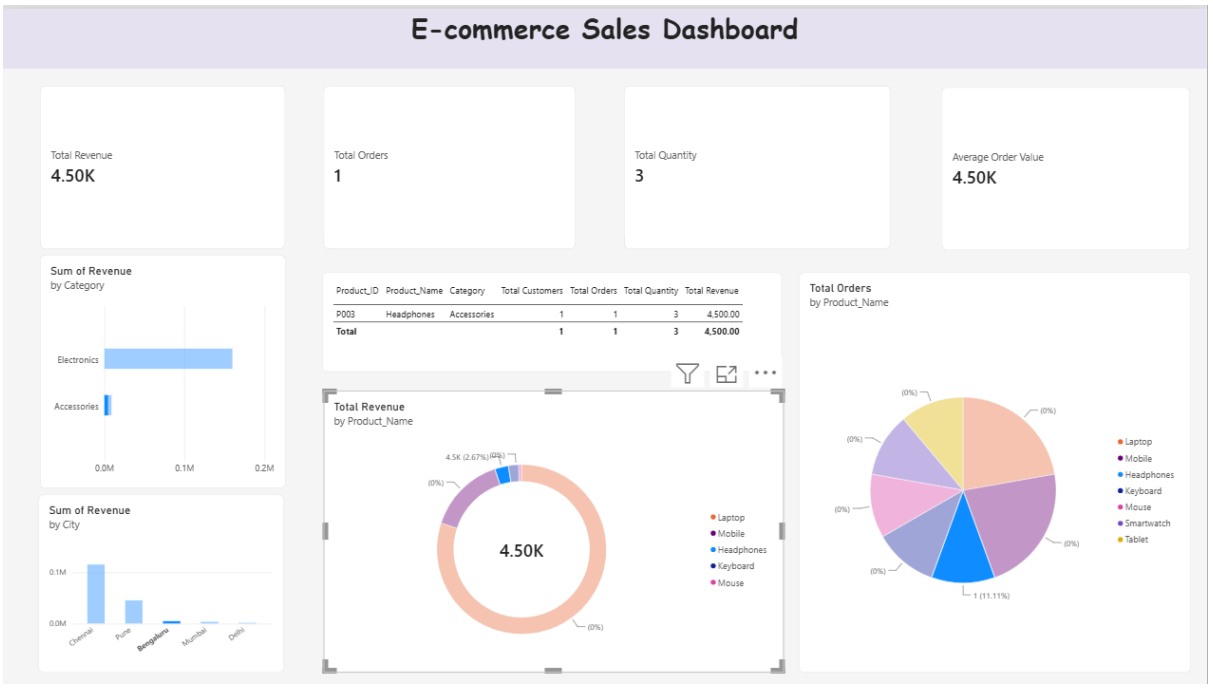

Untitled - Power BI Desktop
Sign in
File
Home
Help
Share
1 Revenue = Fact_Orders[Quantity] * Fact_Orders[Unit_Price]
Dim_Products
Category
Product_ID
Product_Name
Dim_Payments
Payment_ID
Payment_Method
Stg_Orders
City
Customer_Name
Order_Date
Order_ID
Payment_Method
Product
Quantity
Unit Price
Fact_Orders
Dim_Customers.Customer_ID
Dim_Date.Date_ID
Dim_Payments.Payment_ID
Dim_Products.Product_ID
Order_ID
Quantity
Revenue
All tables
Properties
General
Name
Revenue
Description
Enter a description
Synonyms
revenue
Display folder
Enter the display folder
Is hidden
No
Data
Tables
Model
Search
Dim_Customers
Dim_Date
Dim_Payments
Dim_Products
Fact_Orders
Dim_Customers.Customer_ID
Dim_Date.Date_ID
Dim_Payments.Payment_ID
Dim_Products.Product_ID
Order_ID
Quantity
Revenue
Unit_Price
Stg_Orders
100%


Untitled - Power BI Desktop
Sign in
File
Home
Help
Share
Paste
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Transform data
Refresh
Manage relationships
New measure
New column
New table
Calculation group
New parameter
Manage roles
View as
Q&A setup
Language
Linguistic schema
Sensitivity
Clipboard
Data
Queries
Relationships
Calculations
Parameters
Security
Q&A
Sensitivity
1 Total Revenue = SUM(Fact_Orders[Revenue])
Dim_Products
Category
Product_ID
Product_Name
Dim_Payments
Payment_ID
Payment_Method
Stg_Orders
City
Customer_Name
Order_Date
Order_ID
Payment_Method
Product
Quantity
Unit Price
Fact_Orders
Dim_Customers.Customer_ID
Dim_Date.Date_ID
Dim_Payments.Payment_ID
Dim_Products.Product_ID
Order_ID
Quantity
Revenue
_Customers
All tables
Properties
General
Name
Total Revenue
Home table
Fact_Orders
Description
Enter a description
Create with Copilot
Content created by AI may not be accurate or appropriate, so review it carefully. Read terms
Synonyms
Data
Tables
Model
View and organize all of the items in your semantic model. Learn how
Search
Columns (8)
Dim_Customers.Customer_ID
Dim_Date.Date_ID
Dim_Payments.Payment_ID
Dim_Products.Product_ID
Order_ID
Quantity
Revenue
Unit_Price
Hierarchies (0)
Measures (1)
Total Revenue
Partitions (1)
Stg_Orders
100%


Untitled - Power BI Desktop
Sign in
File
Home
Help
Share
Paste
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Transform data
Refresh
Manage relationships
New measure
New column
New table
Calculation group
New parameter
Manage roles
View as
Q&A setup
Language
Linguistic schema
Sensitivity
Clipboard
Data
Queries
Relationships
Calculations
Parameters
Security
Q&A
Sensitivity
1 Total Orders = DISTINCTCOUNT(Fact_Orders[Order_ID])
Dim_Products
Category
Product_ID
Product_Name
Dim_Payments
Payment_ID
Payment_Method
Stg_Orders
City
Customer_Name
Order_Date
Order_ID
Payment_Method
Product
Quantity
Unit Price
Fact_Orders
Dim_Customers.Customer_ID
Dim_Date.Date_ID
Dim_Payments.Payment_ID
Dim_Products.Product_ID
Order_ID
Quantity
Revenue
_Customers
All tables
Properties
General
Name
Total Orders
Home table
Fact_Orders
Description
Enter a description
Create with Copilot
Content created by AI may not be accurate or appropriate, so review it carefully. Read terms
Synonyms
Data
Tables
Model
View and organize all of the items in your semantic model. Learn how
Search
Fact_Orders
Columns (8)
Dim_Customers.Customer_ID
Dim_Date.Date_ID
Dim_Payments.Payment_ID
Dim_Products.Product_ID
Order_ID
Quantity
Revenue
Unit_Price
Hierarchies (0)
Measures (2)
Total Orders
Total Revenue
100%


Untitled - Power BI Desktop
Sign in
File
Home
Help
Share
Paste
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Transform data
Refresh
Manage relationships
New measure
New column
New table
Calculation group
New parameter
Manage roles
View as
Q&A setup
Language
Linguistic schema
Sensitivity
Clipboard
Data
Queries
Relationships
Calculations
Parameters
Security
Q&A
Sensitivity
1 Total Customers = DISTINCTCOUNT(Fact_Orders[Dim_Customers.Customer_ID])
Dim_Products
Category
Product_ID
Product_Name
Dim_Payments
Payment_ID
Payment_Method
Stg_Orders
City
Customer_Name
Order_Date
Order_ID
Payment_Method
Product
Quantity
Unit Price
Fact_Orders
Dim_Customers.Customer_ID
Dim_Date.Date_ID
Dim_Payments.Payment_ID
Dim_Products.Product_ID
Order_ID
Quantity
Revenue
_Customers
All tables
Properties
General
Name
Total Customers
Home table
Fact_Orders
Description
Enter a description
Create with Copilot
Content created by AI may not be accurate or appropriate, so review it carefully. Read terms
Synonyms
Data
Tables
Model
View and organize all of the items in your semantic model. Learn how
Search
Dim_Date.Date_ID
Dim_Payments.Payment_ID
Dim_Products.Product_ID
Order_ID
Quantity
Revenue
Unit_Price
Hierarchies (0)
Measures (3)
Total Customers
Total Orders
Total Revenue
Partitions (1)
Stg_Orders
100%


Untitled - Power BI Desktop
Sign in
File
Home
Help
Share
Paste
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Transform data
Refresh
Manage relationships
New measure
New column
New table
Calculation group
New parameter
Manage roles
View as
Q&A setup
Language
Linguistic schema
Sensitivity
Clipboard
Data
Queries
Relationships
Calculations
Parameters
Security
Q&A
Sensitivity
1 Average Order Value = DIVIDE([Total Revenue],[Total Orders],0)
Dim_Products
Category
Product_ID
Product_Name
Dim_Payments
Payment_ID
Payment_Method
Stg_Orders
City
Customer_Name
Order_Date
Order_ID
Payment_Method
Product
Quantity
Unit Price
Fact_Orders
Dim_Customers.Customer_ID
Dim_Date.Date_ID
Dim_Payments.Payment_ID
Dim_Products.Product_ID
Order_ID
Quantity
Revenue
Dim_Customers
All tables
Properties
General
Name
Average Order Value
Home table
Fact_Orders
Description
Enter a description
Create with Copilot
Content created by AI may not be accurate or appropriate, so review it carefully. Read terms
Synonyms
Data
Tables
Model
View and organize all of the items in your semantic model. Learn how
Search
Dim_Products.Product_ID
Order_ID
Quantity
Revenue
Unit_Price
Hierarchies (0)
Measures (5)
Average Order Value
Total Customers
Total Orders
Total Quantity
Total Revenue
Partitions (1)
Stg_Orders
100%


E-commerce Sales Dashboard
Total Revenue
4.50K
Total Orders
1
Total Quantity
3
Average Order Value
4.50K
Sum of Revenue
by Category
Electronics
Accessories
0.0M
0.1M
0.2M
Product_ID Product_Name Category Total Customers Total Orders Total Quantity Total Revenue
P003 Headphones Accessories 1 1 3 4,500.00
Total 1 1 3 4,500.00
Total Orders
by Product_Name
Sum of Revenue
by City
Chennai
Pune
Bengaluru
Mumbai
Delhi
Total Revenue
by Product_Name
4.50K
Laptop
Mobile
Headphones
Keyboard
Mouse
Smartwatch
Tablet